DIVYA · M · K
20241CSE0568

DAY-1

Date: 01/06/2026

**My objectives for the day:**

→ To Understand the structure and objectives of the AI Summer Intership program. To learn about ethical considerations in AI development and deployment.

**List of activities I did today:**

→ learn about Real-world applications of AI in healthcare, agriculture, education, finance and autonomous systems.
→ Installed and configured python and necessary development libraries
→ Installed and verified the CUDA environment for NVIDIA GPU support
→ Tested the setup by running sample python commands and verifying CUDA availability.

**My critical observation:**

→ AI is transforming multiple industries and creating innovative solutions to complex problems
→ GPU acceleration significantly improves computational performance compared to CPU-only execution.

**What did I learn today:**

→ Current trends and future opportunities in AI.
→ importance of data privacy, fairness and transparency in AI systems
→ Basic of CUDA and GPU computing.

**What I would have done better:**

→ prepared the required software and dependencies before the session to reduce setup time
→ Explored additional documentation on CUDA and GPU architecture.

**My plan for tomorrow:**

→ Review today's concepts related to AI fundamentals and ethics
→ Explore and verify that the development environment is functioning correctly

Student Signature: Divya

Faculty-in-Charge Signature:

DAy-2

Date: 02/06/2026

**My objectives for the day:**

→ To Understand the fundamental concepts of Machine learning
→ To learn the difference b/w Regression and classification problems

**List of activities I did today:**

→ Explored logistic Regression models for classification tasks such as Survival prediction and P/F Prediction.
→ learned the concepts of training datasets, testing datasets, and train - test Splitting.
→ Analyzed confusion matrix to evaluate classification Model accuracy
→ learned about GPU-Supported ML workflows and the advantages of accelerated computation.

**My critical observation:**

→ Proper data preprocessing significantly impacts model performance.
→ GPU acceleration can reduce training time and improve computational efficiency for larger datasets.

**What did I learn today:**

→ Dataset preparation, feature selection and model training
→ pratical implementation of predictive models using python and Sckit - learn.

**What I would have done better:**

→ Explore additional Machine learning datasets for practice
→ praticed implementing Machine learning pipelines independently

**My plan for tomorrow:**

→ Continue hands-on experimentation with datasets and performance evaluation
→ learn more about model optimization techniques

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DAY - 3

Date: 03/06/2026

**My objectives for the day:**

→ learn neural networks and backpropagation.
→ Understand PyTorch basics on H200 GPU

**List of activities I did today:**

→ Studied neural networks from scratch.
→ learned backpropagation concepts
→ Explored PyTorch fundamentals
→ Implemented simple neural network models
→ Executed training on GPU

**My critical observation:**

→ Neural networks learn by adjusting weights through backpropagation
→ GPUs significantly accelerate training

**What did I learn today:**

→ Neural network architecture
→ Backpropagation process
→ Basics of PyTorch framework.

**What I would have done better:**

→ Practiced more PyTorch coding examples
→ Explored more advanced neural network architectures
→ performed additional experiments with different learning rates

**My plan for tomorrow:**

→ learn Convolutional Neural Networks (CNNs)
→ perform image classification experiments

Student Signature: Divyak

Faculty-in-Charge Signature: Atheas.

DAy-4

Date: 04/06/2026

**My objectives for the day:**

→ Understand Convolutional Neural Networks
→ learn image classification techniques
→ Explore GPU profiling tools.

**List of activities I did today:**

→ Attended a lecture on CNN architecture.
→ learned about convolution, pooling, and feature extraction.
→ Studied image datasets and preprocessing techniques.
→ Implemented CNN models using pytorch
→ performed image classification experiments

**My critical observation:**

→ CNNs are highly effective for computer vision applications
→ GPU profiling helps identify performance bottlenecks.

**What did I learn today:**

→ CNN architecture and working principles.
→ GPU profiling and optimization basics
→ importance of data preprocessing

**What I would have done better:**

→ Experimented with deeper CNN architectures.
→ Explored additional optimization methods.

**My plan for tomorrow:**

→ ~~part~~ prepare for the AI fundaments Quiz
→ clarify doubts with mentors before moving to advanced AI topics.

Student Signature: ............................... Faculty-in-Charge Signature: ...............................

Day-5

DIVYA.M.K
20241CSE0568

Date: 05/06/2026

**My objectives for the day:**

→ Explore creative applications of web Technologies
→ improve problem-solving and analytical skills

**List of activities I did today:**

→ Explored interactive web-based simulations and demonstrations.
→ learned how creativity and technology can be combined to enhance user experience.
→ Engaged in discussions about innovative applications of AI and modern technologies.

**My critical observation:**

→ interactive technologies make learning more engaging and enjoyable.
→ continuous revision is essential for retaining technical knowledge

**What did I learn today:**

→ Techniques for improving problem-solving and logical thinking skills.
→ significance of revision and self-assessment in learning.

**What I would have done better:**

→ participated more actively in discussions and practical activities.

**My plan for tomorrow:**

→ Learn about Transformer Architecture and Attention mechanisms
→ prepare for advanced AI and Natural language processing concepts.

Student Signature: Divyak

Faculty-in-Charge Signature:

DAY-6

Date: 08/06/2026

**My objectives for the day:**

→ Advance AI, LLM's & Capstone project
→ Transformer Architecture deep dive / Attention mechanisms BERT & GPT Anatomy

**List of activities I did today:**

On day 6, I explored the architecture of transformer model which forms the best system such as BERT & GPT. In this sessions focused on understanding how transformer process sequential data using attention mechanism instead of traditional neural network.

**My critical observation:**

Self-attention help the model focus on import words in sequence Training loss decreased over epochs, showing effective model learning

**What did I learn today:**

→ Input Embeddings
→ Self-Attention Mechanism
→ Transform traning process
→ GPT ~~to~~ style casual masking

**What I would have done better:**

→ Explored the mathematical calculation behind attention score in more detail visualized attention weights to better understand token relationship

**My plan for tomorrow:**

→ LLM fine-Tuning with LORA & PEFT
→ Hugging Face on H200
→ Prompt engineering

Student Signature: Divyaki

Faculty-in-Charge Signature: [signature]

DAY-7

Date: 09/06/2026

**My objectives for the day:**

→ LLM fine-Tuning with LoRA & PEFT
→ Hugging face on H200
→ prompt engineering

**List of activities I did today:**

→ Studied the fundamentals of large language Model
→ learned about LoRA for efficient modal fine-tuning
→ explored PEFT Techniques
→ worked with Hugging face ecosystem & transformer model
→ Practised prompt engineering techniques.
→ Studied methods to optimize model performance while reducing computational cost

**My critical observation:**

→ LoRA significantly Reduces number of trainable parameters PEFT enables efficient fine-tuning with lower computational cost.

**What did I learn today:**

prompt engineering plays vital role, it depends on How we give prompt for excepting answer.

**What I would have done better:**

Tested different prompt engineering greatly improve AI and response PEFT techniques behind LoRA

**My plan for tomorrow:**

Genrative AI: Diffusion Model & GANS
Stable Diffusion on H200 GPU

Student Signature: ..............................

Faculty-in-Charge Signature: ..............................

DAY- 8

Date: 10/06/2026

**My objectives for the day:**

→ Generative OF AI
→ Diffusion Models & GAN's
→ Static Diffusion on How GPU

**List of activities I did today:**

→ Attended the lecture on Generative AI concepts
→ learned the difference b/w Discriminative AI ~~Gener~~ and Generative AI
→ Studied GANs including Generator and Discrimi-nator architecture

**My critical observation:**

→ GANs can generate image very quickly but are difficult to train because of instability and mode collapse issues.

**What did I learn today:**

→ Generative AI and differences b/w other AI
→ GAN & How does it work.

**What I would have done better:**

→ Spend more time understanding the mathmatical concepts behind diffusion process

**My plan for tomorrow:**

→ learn GPU optimization techniques such as Mixed Precision Training

Student Signature: Divyah

Faculty-in-Charge Signature: Afhan

Day-9

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Date: 11/06/2026

**My objectives for the day:**

-> GPU optimization: Mixed Precision
-> Flash attention, Tensor parallism
Capstone work.

**List of activities I did today:**

- learned GPU optimization techniques like mixed precision and tensor parallelism
- Quiz from (Day 01 to Day 09)
- Pratise of CV problems & yolo

**My critical observation:**

- I could not answer question (during Quiz)

**What did I learn today:**

yolo (helps in OpenCV)

**What I would have done better:**

**My plan for tomorrow:**

Capstone Project presentation
peer review
certificate
closing cermony

Student Signature: Divya

Faculty-in-Charge Signature: Afshan

DAY-10

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Date: 12/06/2026

**My objectives for the day:**

-> GPU Optimization : Mixed Precision
-> flash attention , Tensor Parallism
-> Capstone work -

**List of activities I did today:**

- Delivered our teams Capstone project presentation and Question from evaluation panel
- Activietly participated in peer review session

**My critical observation:**

I noticed that teams who focused heavily on pratical implementation and solve problem solving metrics tended to handle better

**What did I learn today:**

learned how to effectively structure and articulate complex technical concepts.

**What I would have done better:**

I could do better by adding features for our project "Crop care AI"

**My plan for tomorrow:**

TO revise all concepts, work one more project

Student Signature: Divya

Faculty-in-Charge Signature: